

AUTONOMOUS REGULATORY AGENT (ARA)

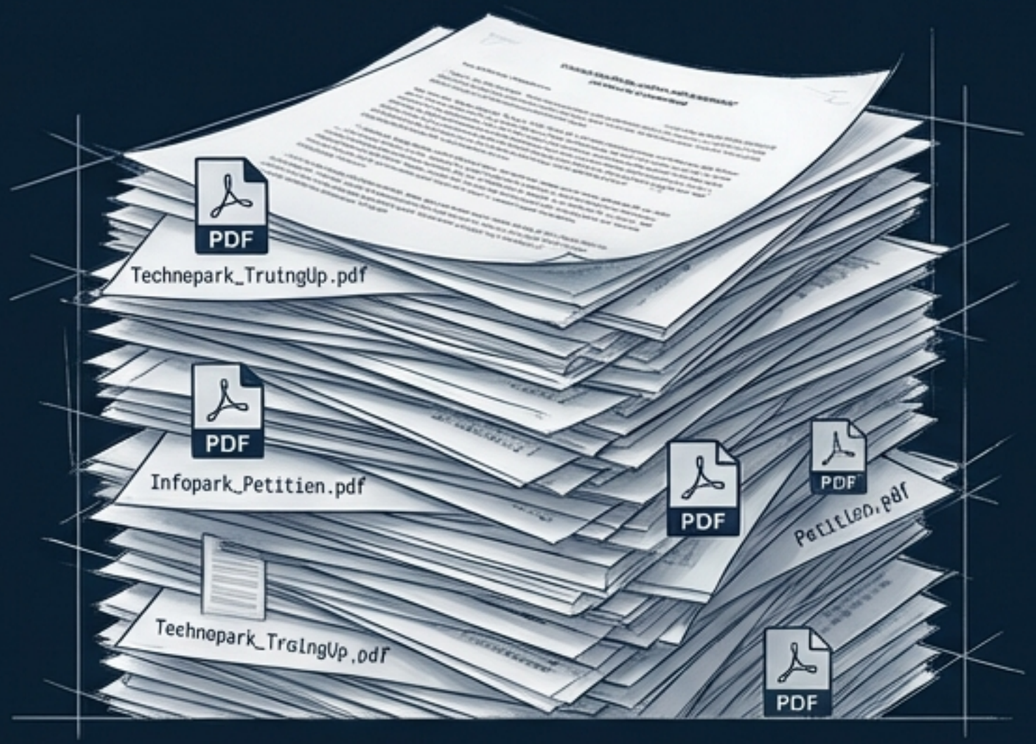
An AI-Based Tool for Smart Truing-Up & Regulatory Compliance.

RAJADHANI INSTITUTE OF ENGINEERING AND TECHNOLOGY (RIET)

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The Problem: Regulatory Complexity & Data Overload

The Input: Unstructured Data



The Need: Mathematical Precision



- **Scope:** Analyzing massive truing-up petitions for licensees like KDHPCL, Technopark, Infopark, and CSEZA.
- **The Burden:** Manual validation of thousands of cost elements (Employee costs, R&M, Power Purchase).
- ⚠ **The Risk:** Missing revenue gaps or approving imprudent expenses due to lack of historical correlation.
- **The Goal:** Automated, mathematical precision in scrutiny using sovereign-grade AI.

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Meet ARA: Your Virtual Regulatory Auditor



Definition: ARA is a custom-built analytical engine, designed specifically for the KSERC workflow.

Role: Embedded within the Commission's workflow to automate the heavy lifting of data extraction and verification.

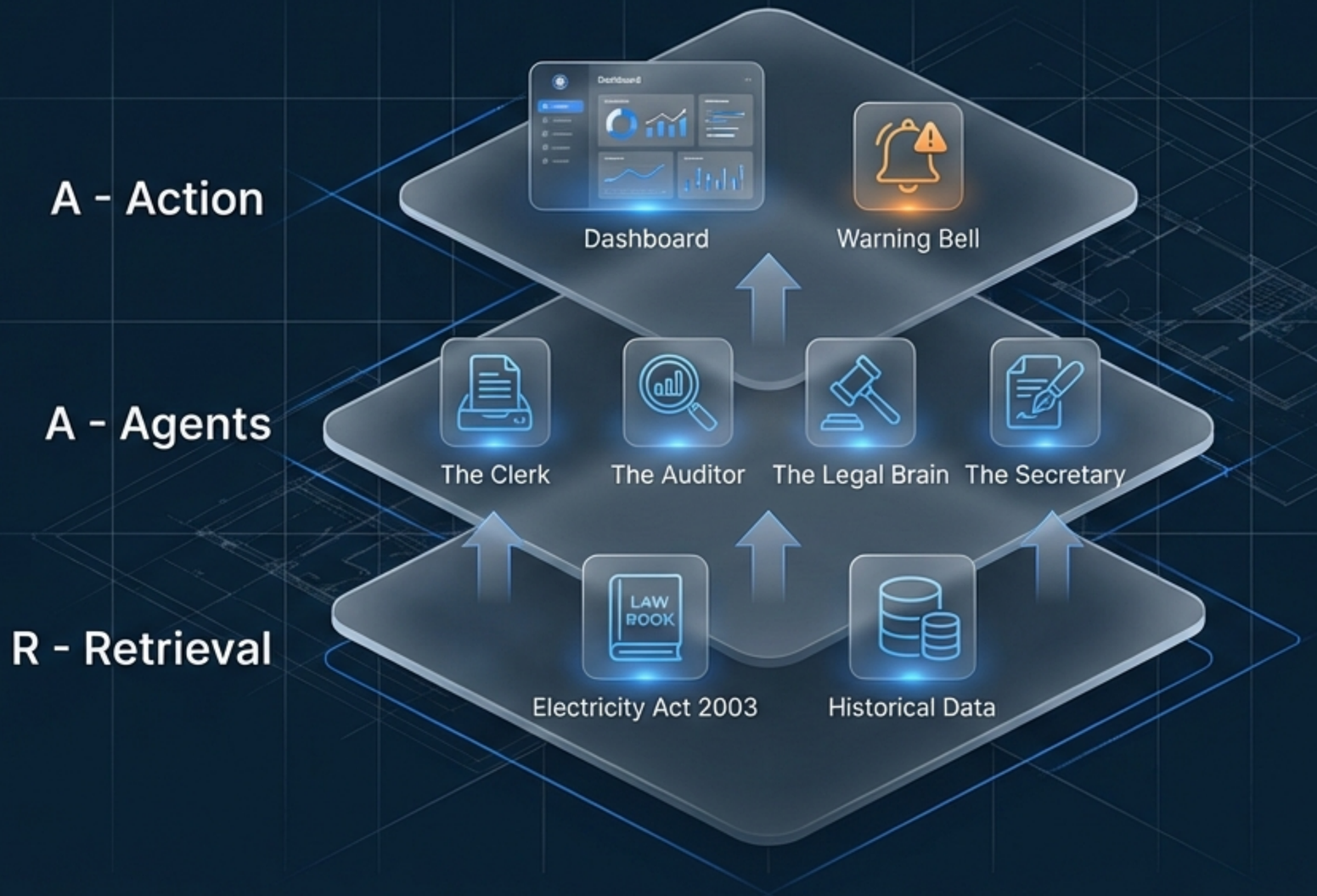
Core Functions:

1. Validates cost elements against approved ARR & ERC norms.
2. Spots revenue gaps (e.g., flagging Technopark's claimed revenue surplus of Rs. 328.09 lakh).
3. Performs prudence checks on Controllable vs. Uncontrollable expenses.

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The ARA Architecture: Agents, Retrieval, Action

Moving from a single model to a Multi-Agent System (MAS).



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The Specialized Workers (Part 1)



Agent 1: The Data Ingestor ("The Clerk")

Purpose: Handles historical and current truing up data.
Action: Pulls data directly from PDF petitions (e.g., Infopark Petition No: OP 59/2025) and maps it to a unified Regulatory Data Schema.



Agent 2: The Trend Analyzer ("The Forensic Auditor")

Purpose: Spots trends, deviations, and inconsistencies.
Real World Trigger: In the 2023-24 Technopark petition, this agent flags an actual distribution loss of 4.06% against the approved 3.35%, triggering an automatic financial disallowance calculation.

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The Specialized Workers (Part 2)



Agent 3: The Compliance Sentinel ('The Legal Brain')

Purpose: Conducts prudence checks based on **Tariff Regulations 2021**.

Action: Distinguishes '**Controllable**' costs (R&M, A&G) vs. '**Uncontrollable**' costs. Validates if expenses like the **Rs. 503.62 lakh** R&M claim followed transparent bidding.



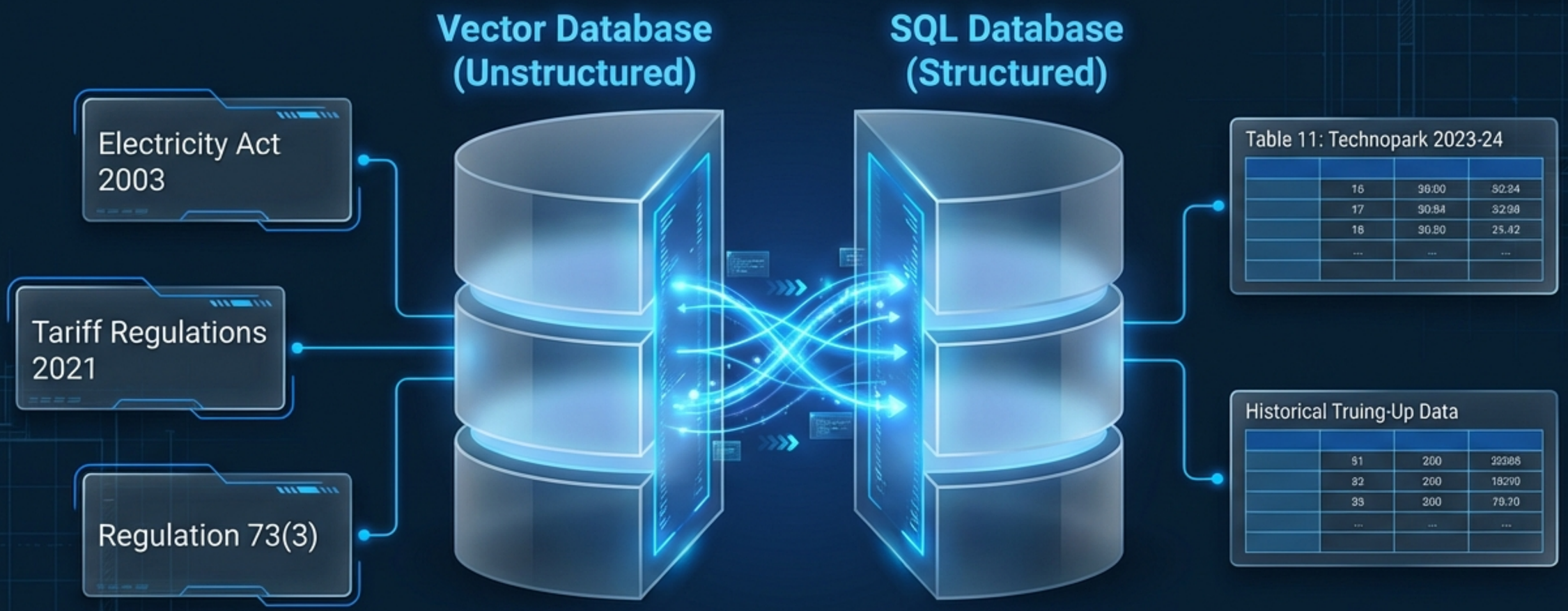
Agent 4: The Reporter ('The Secretary')

Purpose: Generates **analytical reports** for the Commission.

Action: Drafts the '**Analysis and Decision**' sections of the **Regulatory Order**, citing specific regulations (e.g., **Regulation 73(3)** regarding distribution loss variations).

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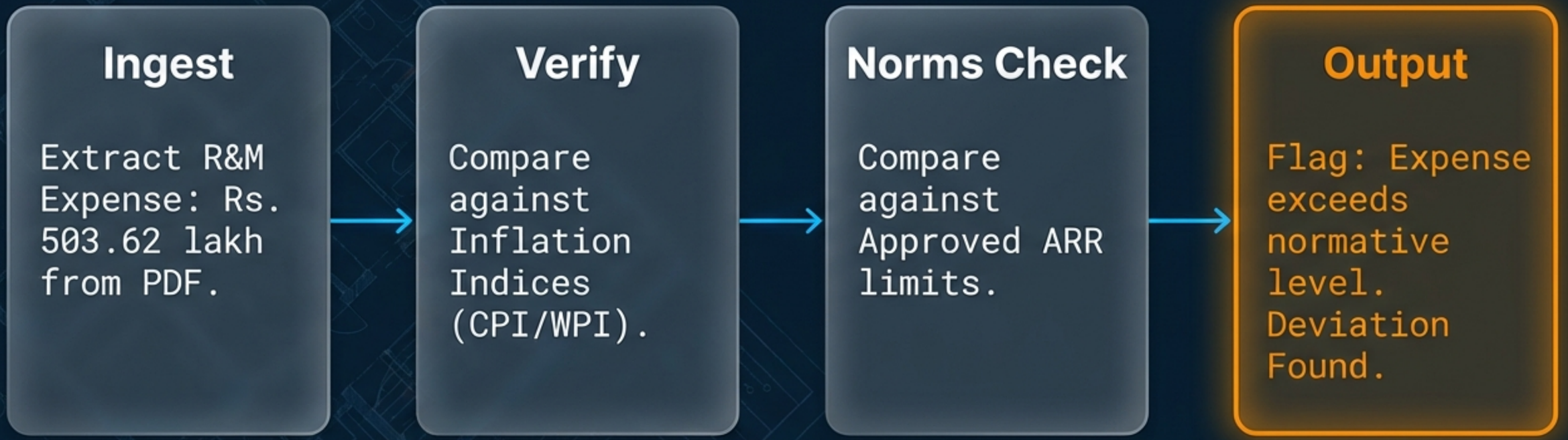
R - Retrieval: A Robust RAG System



The Synergy: The system creates a bridge, allowing it to check if a specific number (Structured) complies with a specific text rule (Unstructured).

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The Brain in Action: Logic Flow



Query: Check Repair & Maintenance Expenses for Prudence.

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Case Study A: The Distribution Loss Trap

Automated Calculation of Disallowances

Scenario: Technopark (2023-24) reports Actual Distribution Loss of 4.06%.

Benchmark: KSERC Approved Loss is 3.35%.

ARA Action: Automatically calculates financial disallowance.

Approved power purchase cost for 2023-24	
Actual distribution loss (In lakh units)	31.51
Approved distribution loss (In lakh units)	25.79
Excess distribution loss (lakh units) (1-2)	5.72
Average Power Purchase Cost (Rs. per unit)	7.19
Excess power purchase cost to be disallowed (Rs. lakh) (4) * (3)	41.13

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Case Study B: The Prudence Check

Context-Aware Analysis of Assets & Expenses



Technical Prudence

Input: Technopark Solar Plant Data.

Observation: Polycrystalline panels showing only 16-17% efficiency; Inverter issues reported.

ARA Flag: Generation significantly lower than Regulation norms. Verify if 'Technical Issues' justify the revenue drop.



Financial Prudence

Input: O&M Expenses (CSEZA/Technopark).

Logic: Categorizes as 'Controllable Expenses'.

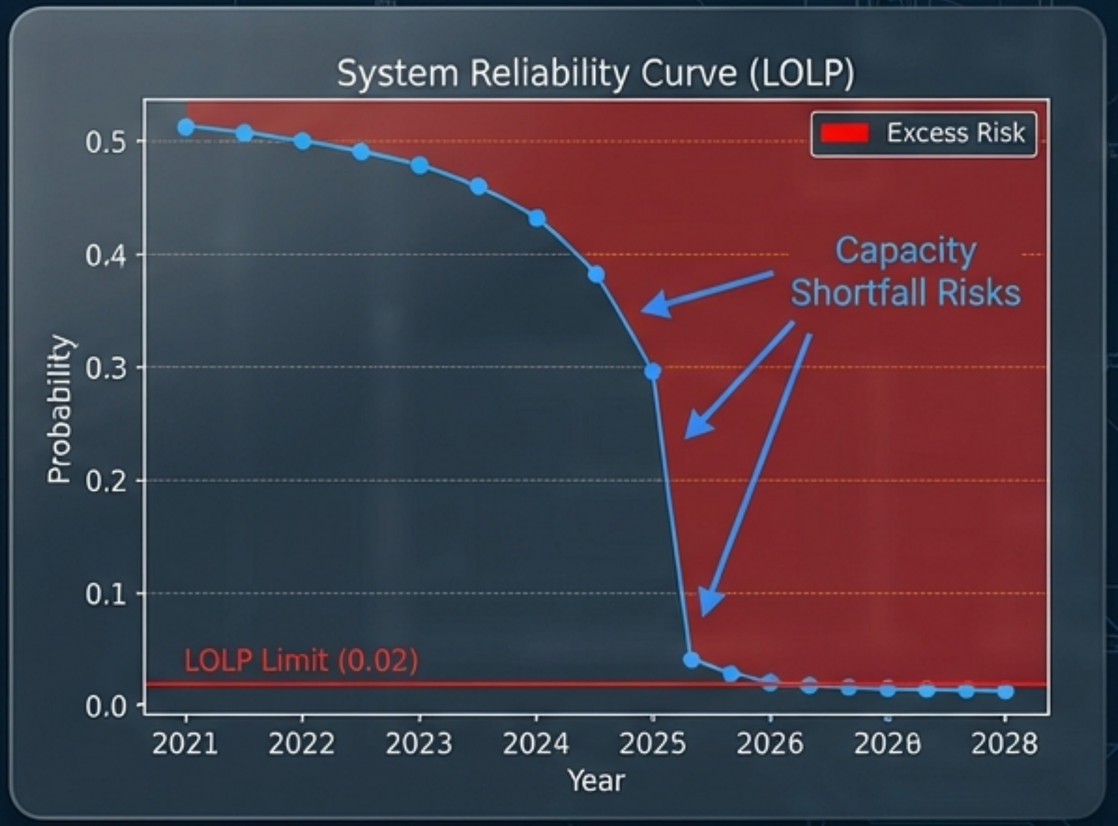
ARA Flag: Expenses exceeding norms must be disallowed unless 'Uncontrollable Factors' (e.g., statutory wage revision) are proven.

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Future Module: Resource Adequacy (RA)

From Accounting to Strategic Planning

- The **New Mandate**: In line with the 2023 Framework, the Commission is moving toward **rigorous generation planning**.
- ARA Capabilities:
  1. **Load Forecasting**: Using ARIMA models to predict demand (MW/MWh) for 10-year horizons.
  2. **Reliability Metrics**: Calculating LOLP (Loss of Load Probability) and EENS.



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Data Confidentiality & Sovereign Security



Local Deployment:

No financial data sent to public cloud APIs.

Confidentiality:

Adheres to Section 5 of the Work Order.

Access Control:

Role-Based Access Control (RBAC) for Commission staff.

IPR:

Vests with the Commission (Work Order Section 6).

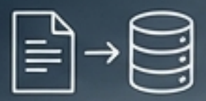
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Implementation Roadmap

Weeks 1-2

Data Ingestion & Schema Design

(Mapping PDF tables to SQL)



Weeks 3-5

Agent Training & Logic Coding

(Encoding Tariff Regs 2021)



Weeks 6-7

UI/Dashboard Development & Report Generation



Week 8

UAT, Validation with Actual Data, and Handover





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The Value Proposition

Manual Scrutiny (Current)

- Time-consuming PDF review.
- Risk of calculation errors.
- Reactive analysis.

ARA (Future)

- **Instant Extraction**: 100+ pages analyzed in minutes.
- **Zero-Error Math**: Automated disallowance calculations (e.g., Rs. 41.13 lakh).
- **Strategic Insight**: Predictive analytics for Resource Adequacy.

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Ready to Deploy



Empowering the Commission with AI-Driven Precision.

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